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Unveiling the projection nature of solvation interactions yields a robust PMPA-MD method for efficient modeling of liquid-phase reactions: A case study on H_2PtCl_6 hydrolysis

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ABSTRACT

While practically essential yet technically challenging, the lack of understanding of the nature of solvation interactions hinders the development of accurate and universal simulation methods for liquid-phase reactions. Here, we introduce an innovative concept of projecting intermolecular solvation interactions onto each atom/species, analogous to the way of quantifying covalent bonding interactions, and thereby developing a powerful tool of the Projective Multi-Point Averaging Molecular Dynamics (PMPA-MD), realizing consistent treatment of diverse solvation scenarios covering both liquid/solid interfaces and homogeneous solution reactions. Taking the multistep H_2PtCl_6 hydrolysis as an example, PMPA-MD demonstrates good accuracy with slight energy deviations <0.1 eV and reduces computational costs by approximately one order of magnitude, as compared with the benchmark constrained MD simulations. We show that the process is typically exothermic (except for the first hydrolysis step) and proceeds with surmountable reaction barriers following a Brønsted–Evans–Polanyi relationship. The substitution site is governed by the *trans* effect, while the substituent group ($-\text{OH}$ or $-\text{H}_2\text{O}$) exhibits piecewise pH dependence. Furthermore, elucidating the dynamic hydrolysis mechanism enables exploring the subsequent process of hydrolysate adsorption and reductive nucleation on the support surface, thereby shedding light on the morphology control of deposited Pt catalysts during experimental synthesis. This work advances both the concept and methodology for liquid-phase studies.

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I. INTRODUCTION

Liquid-phase reactions are fundamentally important across diverse domains, including photo-/electro-catalysis, catalyst synthesis, and biochemical processes. One prominent example is the synthesis of supported Pt catalysts, such as Pt/CeO_2 ^{1,2} and Pt/TiO_2 ,^{3,4} by diluting chloroplatinic acid (H_2PtCl_6) into aqueous solution. Obviously, the hydrolysis of H_2PtCl_6 into $[\text{PtCl}_{6-x}(\text{OH})_x(\text{H}_2\text{O})_{x-y}]^{2-x-y}$ species^{5,6} largely determines the morphology and properties of the deposited Pt catalysts.⁷ The process was typically controlled by altering experimental conditions, such as pH, reactant concentrations,

and temperature, leading to noticeable impacts on the catalytic efficiency and stability of as-synthesized Pt/support catalysts. For example, Toukabri *et al.*⁸ realized the controllable deposition of Pt single atoms or nanoparticles on TiO_2 by adjusting the solution pH. Lv *et al.*⁹ determined a strong correlation between the photodegradation activity of Pt/TiO_2 and the concentration of H_2PtCl_6 precursor for synthesizing the catalyst. Therefore, achieving mechanistic understandings of H_2PtCl_6 hydrolysis is highly desired, as these insights facilitate precise engineering of supported Pt catalysts with optimized structural and catalytic characteristics.

It is known that a fundamental distinction between liquid- and gas-phase reactions arises from the solvation effect,¹⁰ a category of non-covalent interactions that significantly affects both mechanistic pathways and reaction kinetics. For instance, Proutiere and Schoenebeck¹¹ discovered the crucial impacts of solvent polarity on the regioselectivity (i.e., C–Cl activation or C–OTf cleavage) in the Suzuki coupling reaction, while Cram *et al.*¹² observed dramatic solvent-dependent reaction rate variations spanning nine orders of magnitude for the racemization of dimethyl sulfoxide. Technically, the early stage of computational chemistry addressed the solvation effect via implicit models that approximate solvents as continuous dielectric media.^{13,14} However, two fundamental limitations have restricted its broader applications: (i) inability to accurately model solvent-mediated processes (e.g., proton transfer), and (ii) inadequate description of site-specific solvation effects. In contrast, the development of explicit solvation methods, such as constrained MD^{15–17} and umbrella sampling,¹⁸ tackled these limitations by explicitly accounting for solvent molecules. For example, Zhao *et al.*¹⁹ investigated the first step of proton-coupled electron transfer in CO₂ reduction on single-nickel atom catalysts and revealed how electric potential directly controls CO₂ conversion steps. Similarly, Qian *et al.*²⁰ conducted constrained MD simulations and pinpointed the formation of high-valent *NO₃^{2–} intermediate as the primary cause of the high overpotential in nitrate electroreduction. Nevertheless, the requirement for whole-process thermodynamic integration (via dense sampling) along the reaction coordinate using explicit solvation *ab initio* MD (AIMD) simulations, while theoretically rigorous, imposes prohibitive computational costs that restrict its practical use.^{21,22}

Alternatively, we previously proposed a Multi-Point Averaging Molecular Dynamics (MPA-MD) method that incorporates solvation interactions while avoiding biased energy contributions from fluctuant solvent configurations for each intermediate state, realizing the state-to-state energetics calculations (instead of whole-process thermodynamic integration) on liquid/solid interface reactions. The method enables precise determination of the point of zero charge (PZC) of TiO₂ and radical-driven photocatalytic oxygen evolution mechanism at water/TiO₂ interfaces.^{23,24} Furthermore, it was successfully extended to the bubble/water/TiO₂ triple-phase interface, revealing how the local hydrogen bond network controls the photocatalytic kinetics of H₂O₂ formation.^{25,26} Note that, while surface-bound species at the liquid/solid interface typically share merged solvation shells with the solid catalyst, homogeneous catalysis involves frequent ligand exchange and, thus, features characteristic transitions between merged and separated solvation states. Obviously, the accurate treatment of solvation interactions in these distinct scenarios is critical for liquid-phase computational methods, although developing a robust and consistently applicable framework for diverse liquid-phase reactions remains a central challenge in computational chemistry.

In this work, we unveiled the projection nature of solvation interactions and thereby developed an accurate and efficient PMPA-MD method for simulating liquid-phase reactions. The reliability of the method was validated through systematic investigations of H₂PtCl₆ hydrolysis as a model system. The manuscript is structured as follows: First, the concepts of the ambient solvation approach via MPA-MD method and the projective solvation approach via PMPA-MD method are presented, where the two approaches show

distinct differences in handling processes involving transitions between merged and separated solvation shells. Second, the efficiency and reliability of PMPA-MD were validated by comparing the determined reaction energies and barriers of three representative steps of H₂PtCl₆ hydrolysis with those from the benchmark constrained MD simulations. Third, the full reaction mechanism of H₂PtCl₆ hydrolysis, involving competitive events of substitution stereoselectivity (*trans* or *cis*) and substituent groups (–H₂O or –OH), was clarified at the atomic level. Finally, the pH-dependent speciation of Pt complexes and its impacts on the morphology control of Pt deposition (particles vs single atoms) on TiO₂ surfaces were discussed.

II. COMPUTATIONAL DETAILS

A. DFT calculations

All the calculations were carried out using the Vienna *ab initio* simulation package (VASP),^{27,28} and the electronic exchange and correlation were described within the generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) functional.²⁹ The core-valence electron interactions were presented by the projector-augmented wave (PAW) method,^{30,31} in which the H 1s, O 2s2p, Cl 3s3p, Ti 3d4s, and Pt 5d6s electrons were explicitly treated as the valence electrons. The valence electronic states were expanded in plane wave basis sets with an energy cutoff of 400 eV for liquid systems and 450 eV for TiO₂ models. The occupancy of the one-electron states was calculated using the Gaussian smearing with SIGMA = 0.05 eV. Structure optimizations were conducted using the quasi-Newton Broyden minimization scheme until the atomic forces on each ion were less than 0.05 eV/Å. The transition states (TSs) were searched using a constrained optimization scheme³² and verified by satisfying two criteria: (i) all forces on atoms vanish, and (ii) the total energy is a maximum along the reaction coordination but a minimum with respect to the remaining degrees of freedom.

Pt complexes adsorbed on the anatase TiO₂(101) surface were simulated using gas/solid models with a large vacuum thickness of ~22 Å between periodic slabs. The anatase TiO₂(101) surface was constructed as a four-TiO₂-layer *p*(2 × 3) slab cell (11.0 × 11.4 × 35 Å³), and a corresponding 1 × 1 × 1 K-point mesh was used throughout the calculations. TiO₂ surface structures, under acidic/alkaline conditions, were represented by introducing H/OH groups on the surface and OH/H counterparts on the opposite slabs, which accordingly present H⁺/OH[–] ions (H → H⁺ + e[–]; OH + e[–] → OH[–]) on the reacting surface in a charge-neutral system. In addition, to model the photoreduction process of Pt(IV) to Pt(II) species, extra electrons were introduced by altering the quantity of H over OH (e.g., adding H or removing OH groups on the opposite slabs). Valence states of Pt complexes were verified by Bader charge analysis.

The adsorption energy (E_{ads}) of Pt complexes was calculated according to the following equation:

$$E_{\text{ads}} = E_{\text{Pt/TiO}_2} - E_{\text{Pt}} - E_{\text{TiO}_2}, \quad (1)$$

where $E_{\text{Pt/TiO}_2}$ and E_{TiO_2} represent the total energy of the TiO₂ slab with and without Pt complexes, respectively, and E_{Pt} represents the Pt complex energy in the gas phase.

The reduction energy (E_{red}) of a Pt(IV) complex to the Pt(II) was calculated as follows:

$$E_{\text{red}} = E_{\text{Pt(II)/TiO}_2} - E_{\text{Pt(IV)/TiO}_2}, \quad (2)$$

where $E_{\text{Pt(IV)/TiO}_2}$ and $E_{\text{Pt(II)/TiO}_2}$ represent the total energy of a six-fold coordination Pt(IV) and a four-fold coordination Pt(II) complex plus two detached ligands adsorbed on the TiO_2 surface, respectively.

B. AIMD simulations

AIMD simulations were performed in a periodic cell ($13.5 \times 15.6 \times 14.7 \text{ \AA}^3$) containing 100 H_2O and 1 H_2PtCl_6 molecules with water density of 1.00 g/cm^3 , and a corresponding $1 \times 1 \times 1$ K-point mesh was used during calculations. The simulation temperature was set at 300 K with a stepwise movement of 0.5 fs in a canonical constant-volume, constant-temperature (NVT) ensemble employing Nosé–Hoover thermostats.³³ Each simulation was continued for an additional 5 ps after the energy fluctuations converged, resulting in a total duration of 10–20 ps. The instant dissociation of H_2PtCl_6 into an $[\text{PtCl}_6]^{2-}$ ion and two H^+ ions (in the form of H_3O_2^+ cations;³⁴ Fig. S1) was observed after 0.5 ps simulations. Note that subsequent hydrolysis steps involve $-\text{Cl}$ ligand substitution by $-\text{H}_2\text{O}$ or $-\text{OH}$ groups with concurrent Cl^- release, and thus, we effectively removed one HCl molecule (a pair of H^+ and Cl^- ions) while introducing one H_2O molecule to the solution for every later hydrolysis step. This approach avoids the complicated interactions arising from excessive Cl^- ions in solution, while remaining physically realistic since released ions become readily dispersed in the bulk solvent. While determining the total energies of the hydrolysis states, the distinct $-\text{OH}/\text{H}_2\text{O}$ coordination forms were fully considered according to their occurrence ratios during MD simulations.

C. Constrained MD

For the hydrolysis of the Pt–Cl bond, the initial bond length was about 2.30–2.35 Å. The Pt–Cl bond distance was increased incrementally by 0.4 Å for configurations away from the TS, and by 0.05–0.2 Å for configurations near the TS region. We sampled >12 configurations along the reaction coordinate for each hydrolysis step and conducted constrained AIMD simulations for all configurations at a constant temperature ($T = 300 \text{ K}$) until the interatomic force (the Helmholtz free-energy gradients $\frac{\partial A}{\partial l}$) between the two constrained atoms converged. Note that two distinct types of TS structures (see Fig. S2 for an example) were identified: (i) a final-state-like (FS-like) TS, characterized by quasi-cleavage of the $\text{Pt} \cdots \text{Cl}$ bond accompanied by pre-formation of a $\text{Pt} \cdots \text{OH}_2$ bond, and (ii) an initial-state-like (IS-like) TS, involving concerted quasi-cleavage of the $\text{Pt} \cdots \text{Cl}$ bond and quasi-formation of a $\text{Pt} \cdots \text{OH}_2$ bond. The potential of mean force was computed by averaging the free-energy gradients of the two scenarios, and the total free-energy change ΔA was then determined by integrating the $\frac{\partial A}{\partial l}$ along the reaction coordinate l ,

$$\Delta A = \int_{l_1}^{l_2} \frac{\partial A}{\partial l} * dl. \quad (3)$$

D. Machine learning force field (MLFF)

The MLFF accelerated MD (MLFF-MD) simulations were conducted using the on-the-fly machine learning method implanted in the VASP software version 6 or later,³⁵ employing the NVT ensemble and Nosé–Hoover thermostat at 300 K. The *train* and *refit* modes were used for obtaining an effective MLFF, while the *run* mode was used for performing MLFF-MD. The MLFF accuracy³⁶ was confirmed via the root mean square error (RMSE) on both the training set and an independent validation set, as well as the deviations of the H–H pair correlation function among ~3 ps period simulations using AIMD and MLFF-MD. To construct the validation set, 500 configurations were uniformly extracted from an additional AIMD trajectory to ensure those configurations were not used in the training process. After validation, the MLFF-MD approach enables dramatic extension of simulation timescales at very low computational cost, facilitating exhaustive sampling of liquid-phase reaction pathways over long durations.

III. RESULTS AND DISCUSSION

A. Development of the PMPA-MD approach

Similar to our previous MPA-MD method,^{23–26} the total energy of liquid-phase systems (E_{tot}) can be divided into three components: target solute species (E_{solute}), solvent shell (E_{solvent}), and the solvation interactions between the two (E_{int}). Owing to the biased energy contributions from fluctuating solvent configurations, the liquid-phase reaction energy should be based on the solute's energy within the solvation environment ($E_{\text{sol-included}}$). This quantity comprises the intrinsic energy of the solute (E_{solute}) and the solvation energy (E_{sol}) acting upon it, while explicitly excluding the energy of the solvent part. Two distinct approaches, conceptualizing solvation interactions as either ambient or projective in nature, were proposed to quantify the key term of E_{sol} :

1. **Ambient solvation approach (MPA-MD method)** that counts in the whole solvation interactions induced by the solvent shell to the solute ($E_{\text{sol}}^{\text{ambient}} = E_{\text{int}}$).
2. **Projective solvation approach (PMPA-MD method)** that projects intermolecular solvation interactions E_{int} onto each atom/species, analogous to the Lewis theory of bonding, in quantifying covalent bonding interactions.

Taking the migration of a H_2O molecule from distant solution to the first solvation shell of a PtCl_6^{2-} ion as an example, the process involves the merging of two individual solvation shells (of the H_2O and PtCl_6^{2-} , respectively) into a unified one around the $\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}$ pair. As illustrated in Fig. 1, while MPA-MD and PMPA-MD divide similar solvation shell boundary for the separated state [Fig. 1(a)], MPA-MD treats the merged $\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}$ pair as a whole [Fig. 1(b)], whereas PMPA-MD projects E_{int} on each species [Fig. 1(c)] as follows:

$$\begin{aligned} E_{\text{sol-included}}^{\text{Ambient}}(\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}) \\ = E_{\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}} + E_{\text{sol}}^{\text{ambient}}(\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}), \end{aligned} \quad (4)$$

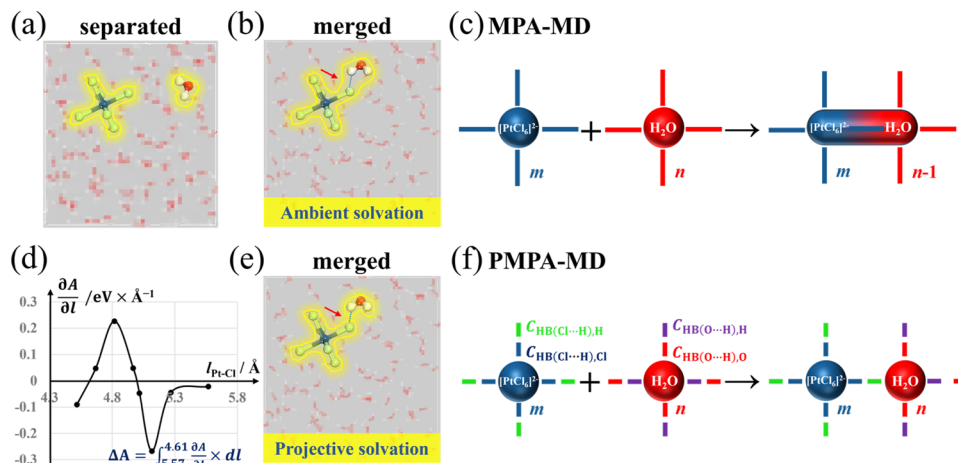


FIG. 1. (a), (b), and (e) Structural snapshots, (c) and (f) schematic reaction diagram, and (d) free-energy gradient profile (via constrained MD simulations) of H_2O migration from distant solution to a PtCl_6^{2-} ion. While the separated solvation shells in (a) are clearly presented by bright yellow margins, the merged solvation states between PtCl_6^{2-} and adjacent H_2O can be described through the (b) ambient or (e) projective solvation approaches using the (c) MPA-MD or (f) PMPA-MD methods. The quantities of hydrogen bonds are indicated as m and n , and the solvation contributions to each species in (f) are presented with projection coefficients as $C_{\text{HB}(\text{Cl}\cdots\text{H})}$, C_{Cl} , $C_{\text{HB}(\text{O}\cdots\text{H})}$, C_{H} , $C_{\text{HB}(\text{O}\cdots\text{H})}$, O , and $C_{\text{HB}(\text{O}\cdots\text{H})}$, H , respectively. Green, red, white, and cyan balls represent Cl, O, H, and Pt atoms, respectively. This color scheme is used throughout the paper.

$$\begin{aligned} E_{\text{sol-included}}^{\text{Projective}}(\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}) \\ = E_{\text{sol-included}}^{\text{Projective}}(\text{H}_2\text{O}) + E_{\text{sol-included}}^{\text{Projective}}(\text{PtCl}_6^{2-}) \\ = E_{\text{H}_2\text{O}} + E_{\text{sol}}^{\text{projective}}(\text{H}_2\text{O}) + E_{\text{PtCl}_6^{2-}} + E_{\text{sol}}^{\text{projective}}(\text{PtCl}_6^{2-}). \end{aligned} \quad (5)$$

These indicate that the ambient solvation model treats the hydrogen bond of the $\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}$ as an internal interaction and counts only external interactions imposed by the water shell as the ambient solvation energy. In contrast, the projective solvation does not differentiate region-dependent (internal vs external) hydrogen bonds and treats all such interactions as projection on relevant atoms according to the respective coefficient. Specifically, the two solute species H_2O and PtCl_6^{2-} provide mutual solvation effects on each other, and the solvent shell offers projected solvation energies to each solute. We assume that the solvation shell induces m and n hydrogen bonds to the PtCl_6^{2-} and H_2O , denoted as $\text{HB}(\text{Cl}\cdots\text{H})$ and $\text{HB}(\text{O}\cdots\text{H})$, respectively. According to the ambient solvation approach [Fig. 1(c)] using the MPA-MD method, the reaction energy $\Delta E_{\text{Ambient}}$ of the process can be deduced as follows:

$$\begin{aligned} \Delta E_{\text{Ambient}} &= E_{\text{sol-included}}^{\text{FS}} - E_{\text{sol-included}}^{\text{IS}} \\ &= [E_{\text{PtCl}_6} + E_{\text{H}_2\text{O}} + m \times E_{\text{HB}(\text{Cl}\cdots\text{H})} + (n-1) \times E_{\text{HB}(\text{O}\cdots\text{H})}] \\ &\quad - [E_{\text{PtCl}_6} + m \times E_{\text{HB}(\text{Cl}\cdots\text{H})} + E_{\text{H}_2\text{O}} + n \times E_{\text{HB}(\text{O}\cdots\text{H})}] \\ &= -E_{\text{HB}(\text{O}\cdots\text{H})}. \end{aligned} \quad (6)$$

Otherwise, by determining approximately the projection coefficient (C_i) as the respective proportion in the sum of van der Waals radius³⁷ between two interacting atoms, giving rise to the $C_{\text{HB}(\text{O}\cdots\text{H})}$ value of 0.54/0.46 for O/H in $\text{HB}(\text{O}\cdots\text{H})$ and $C_{\text{HB}(\text{Cl}\cdots\text{H})}$ of 0.60/0.40 for Cl/H in $\text{HB}(\text{Cl}\cdots\text{H})$, $\Delta E_{\text{Projective}}$ can be deduced according to the projective solvation approach [Fig. 1(f)] using the PMPA-MD method as follows:

$$\begin{aligned} E_{\text{sol-included}}^{\text{IS}} &= E_{\text{sol-included}}^{\text{PtCl}_6} + E_{\text{sol-included}}^{\text{H}_2\text{O}} \\ &= \left(E_{\text{PtCl}_6} + \sum_1^m C_{\text{HB}(\text{Cl}\cdots\text{H})} E_{\text{HB}(\text{Cl}\cdots\text{H})} \right) \\ &\quad + \left(E_{\text{H}_2\text{O}} + \sum_1^n C_{\text{HB}(\text{O}\cdots\text{H})} E_{\text{HB}(\text{O}\cdots\text{H})} \right) \\ &= (E_{\text{PtCl}_6} + m \times 0.6 E_{\text{HB}(\text{Cl}\cdots\text{H})}) \\ &\quad + \left(E_{\text{H}_2\text{O}} + \sum_1^n C_{\text{HB}(\text{O}\cdots\text{H})} E_{\text{HB}(\text{O}\cdots\text{H})} \right) \\ &= E_{\text{PtCl}_6} + E_{\text{H}_2\text{O}} + m \times 0.6 E_{\text{HB}(\text{Cl}\cdots\text{H})} \\ &\quad + \sum_1^n C_{\text{HB}(\text{O}\cdots\text{H})} E_{\text{HB}(\text{O}\cdots\text{H})}, \end{aligned} \quad (7)$$

$$\begin{aligned} E_{\text{sol-included}}^{\text{FS}} &= E_{\text{sol-included}}^{\text{PtCl}_6} + E_{\text{sol-included}}^{\text{H}_2\text{O}} \\ &= (E_{\text{PtCl}_6} + m \times 0.6 E_{\text{HB}(\text{Cl}\cdots\text{H})}) + \left[E_{\text{H}_2\text{O}} + 0.4 E_{\text{HB}(\text{Cl}\cdots\text{H})} \right. \\ &\quad \left. + \left(\sum_1^n C_{\text{HB}(\text{O}\cdots\text{H})} E_{\text{HB}(\text{O}\cdots\text{H})} - 0.46 E_{\text{HB}(\text{O}\cdots\text{H})} \right) \right] \\ &= E_{\text{PtCl}_6} + E_{\text{H}_2\text{O}} + (m \times 0.6 + 0.4) E_{\text{HB}(\text{Cl}\cdots\text{H})} \\ &\quad + \left(\sum_1^n C_{\text{HB}(\text{O}\cdots\text{H})} E_{\text{HB}(\text{O}\cdots\text{H})} - 0.46 E_{\text{HB}(\text{O}\cdots\text{H})} \right), \end{aligned} \quad (8)$$

$$\begin{aligned} \Delta E_{\text{Projective}} &= E_{\text{sol-included}}^{\text{FS}} - E_{\text{sol-included}}^{\text{IS}} \\ &= 0.4 E_{\text{HB}(\text{Cl}\cdots\text{H})} - 0.46 E_{\text{HB}(\text{O}\cdots\text{H})}, \end{aligned} \quad (9)$$

in which $E_{\text{HB}(\text{O}\cdots\text{H})}$ and $E_{\text{HB}(\text{Cl}\cdots\text{H})}$ represent the interaction energy of one $\text{H}_2\text{O} \cdots \text{H}_2\text{O}$ and one $\text{H}_2\text{O} \cdots \text{PtCl}_6^{2-}$ hydrogen bond, respectively.

TABLE I. Calculated energy changes ($\Delta E_{\text{Ambient}}$, $\Delta E_{\text{Projective}}$, ΔA , and zero-point energy and entropy $\Delta \text{ZPE}-T\Delta S$) of H_2O migration from distant solution to the first solvation shell of a PtCl_6^{2-} via the MPA-MD, PMPA-MD (both standard and simplified procedures), and the benchmark constrained MD methods.

Energy change (eV)	$\Delta E_{\text{Ambient}}$ (MPA-MD)	$\Delta E_{\text{Projective}}$ (PMPA-MD)		$\Delta \text{ZPE}-T\Delta S$	ΔA
		Standard	Simplified		
H_2O migration	0.43	0.07	0.08	-0.03	0.01

Furthermore, by computing the ambient solvation energies of H_2O ($E_{\text{sol}}^{\text{ambient}} = E_{\text{tot}} - E_{\text{solute}} - E_{\text{solvent}} = E_{\text{int}}$) and statistically analyzing hydrogen bond numbers m in AIMD trajectories, $E_{\text{HB}(\text{O}\cdots\text{H})}$ can be readily determined as $E_{\text{sol}}^{\text{ambient}}(\text{H}_2\text{O})/m$, being around -0.40 eV. Similarly, $E_{\text{HB}(\text{Cl}\cdots\text{H})}$ was determined to be slightly weaker than $E_{\text{HB}(\text{O}\cdots\text{H})}$, being around -0.31 eV. Table I presents the reaction energies determined via the MPA-MD, PMPA-MD, and the benchmark constrained MD method ($\Delta E_{\text{Ambient}}$, $\Delta E_{\text{Projective}}$, and ΔA). Noticeably, the determined MPA-MD and PMPA-MD results consist well with the derived equations [Eqs. (6) and (9)] by substituting $E_{\text{HB}(\text{O}\cdots\text{H})}$ and $E_{\text{HB}(\text{Cl}\cdots\text{H})}$ values (0.43 vs 0.40 eV; 0.07/0.08 vs 0.06 eV), manifesting the physical validation and reliability of the two methods. Moreover, one can see minor deviations of 0.06/0.07 eV in PMPA-MD, whereas a considerable discrepancy of 0.42 eV for MPA-MD, relative to the constrained MD results. Given the negligible contribution from entropy and zero-point-energy corrections ($\Delta \text{ZPE}-T\Delta S$ in Table I),^{23–26} such contrast credits the projection nature of solvation interactions and the reliability of PMPA-MD method for handling processes involving transitions between merged and separated solvation states, e.g., a typical ligand exchange scenario in homogeneous catalysis.^{38,39}

However, as one may notice, the technical complexity of PMPA-MD would rise significantly for handling mix-ligand species, e.g., intermediate hydrolysates $[\text{PtCl}_{6-x}(\text{OH})_y(\text{H}_2\text{O})_{x-y}]^{-2+x-y}$, since for each type i of related hydrogen bonds around j relevant solutes, the E_{HB}^i , C_i , and quantity N_i need to be subtly determined as follows:

$$E_{\text{sol}}^{\text{projective}} = \sum_1^j \left(\sum_1^i N_i C_i E_{\text{HB}}^i \right). \quad (10)$$

Encouragingly, as inspired by the Lewis theory of bonding,^{40,41} where each atom attains a stable octet configuration by receiving shared electrons via forming covalent bonds, we proposed to

project the whole solvation interaction onto each interacting atom (i.e., C_i is approximate to 1). Such technical operation simplifies the PMPA-MD procedure remarkably as follows:

$$E_{\text{sol-simp}}^{\text{projective}} = \sum_1^j \left(\sum_1^i N_i E_{\text{HB}}^i \right) = \sum_1^j (E_{\text{tot}} - E_{\text{solute}} - E_{\text{solvent}}), \quad (11)$$

and yields results identical to MPA-MD for states with separated solvation shells but distinctively different for those of merged states [Figs. 1(b) and 1(e) and Eqs. (4) and (5)]. Table I demonstrated that simplified PMPA-MD achieves good accuracy with comparable $\Delta E_{\text{Projective}}$ of 0.08 eV relative to that of 0.01 eV via the constrained MD for H_2O migration. Hence, we employed the simplified PMPA-MD approach for the subsequent calculations on H_2PtCl_6 hydrolysis, and its reliability is consistently demonstrated in Table II.

B. Reliability validation

As the whole H_2PtCl_6 hydrolysis process involves the dechlorination of both the Cl-rich and $\text{H}_2\text{O}/\text{OH}$ -rich Pt complexes, we considered the first, second, and fourth dechlorination steps of H_2PtCl_6 hydrolysis, denoted as 1Cl-Hydro., *cis*-2Cl-Hydro. (*cis*- represents the stereoisomer of hydrolysates), and *cis*-4Cl-Hydro., respectively, to further verify the reliability of the projective solvation approach, via extensive AIMD-based simulations using the MPA-MD, PMPA-MD, and constrained MD methods. Taking the 1Cl-Hydro. as an example (Fig. 2), the hydrolysis of a Pt-Cl bond involves the gradual elongation of the Pt-Cl bond under the accommodation of adjacent water molecules and subsequently associates with the formation of a Pt-OH₂ (or Pt-OH) bond. By employing the Pt-Cl distance (l) as the reaction coordinate, the constrained MD simulation determined the whole $\frac{\partial A}{\partial l}-l$ curve, which shows four intersections ($\frac{\partial A}{\partial l} = 0$) corresponding to the IS, TS, and two FSs featuring with (FS') or without (FS'') shared hydrogen bonds between

TABLE II. Calculated energy changes ($\Delta E_{\text{Ambient}}$, $\Delta E_{\text{Projective}}$, ΔA , and $\Delta \text{ZPE}-T\Delta S$) of three hydrolysis steps of 1Cl-Hydro., *cis*-2Cl-Hydro. and *cis*-4Cl-Hydro. via the MPA-MD, PMPA-MD, and benchmark constrained MD methods.

Energy change (eV)	$\Delta E_{\text{Ambient}}$ (MPA-MD)	$\Delta E_{\text{Projective}}$ (PMPA-MD)	$\Delta \text{ZPE}-T\Delta S$	ΔA
1Cl-Hydro.	-0.22	0.11	-0.01	0.01
<i>cis</i> -2Cl-Hydro.	-1.16	-0.67	-0.03	-0.63
<i>cis</i> -2Cl-Hydro.-TS	...	0.77	...	0.70
<i>cis</i> -4Cl-Hydro.	-1.17	-0.76	0.02	-0.71

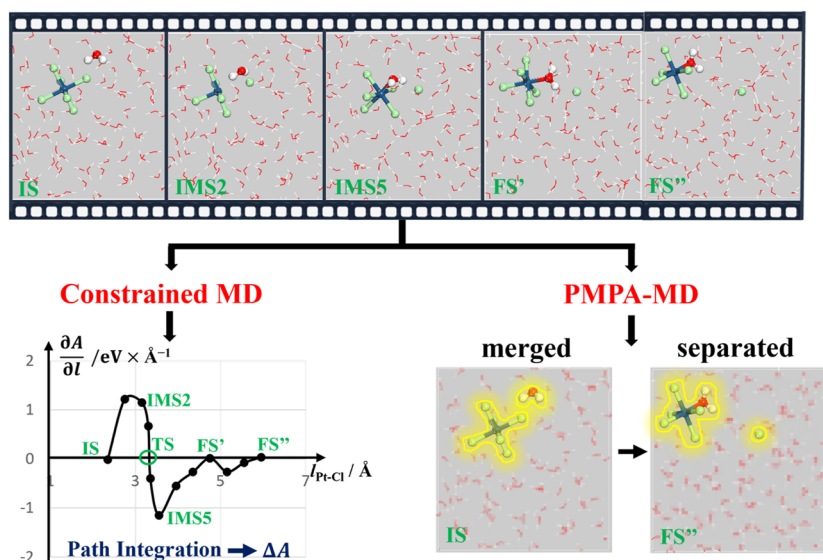


FIG. 2. Dynamic structural changes (top panel) and free-energy gradient profile (via constrained MD simulations; bottom left) of the 1Cl-Hydro. step in H_2PtCl_6 hydrolysis. The two types of structures (initial-state-like and final-state-like) of the TS are provided in Fig. S2. The transition from a merged solvation shell in IS to a separated one in FS'' (bottom right) via the PMPA-MD calculations was depicted with bright yellow margins.

the hydrolysate and the detached Cl^- ion, respectively. The free-energy change ΔA of the process ($\text{IS} \rightarrow \text{FS}''$) can be readily obtained by integrating $\frac{\partial A}{\partial l}$ along the l coordinate using the constrained MD, or by incorporating the $\Delta\text{ZPE-T}\Delta\text{S}$ corrections on the $\Delta E_{\text{Projective}}$ and $\Delta E_{\text{Ambient}}$ using the PMPA-MD and MPA-MD methods, respectively.

Table II shows the determined reaction energies via the three methods. PMPA-MD achieves consistently remarkable agreement with the benchmark constrained MD results (see calculation information in Fig. S3), showing a small energy difference within 0.1 eV for complicated liquid-phase reactions. Moreover, PMPA-MD also allows for the computation of the reaction barrier accurately. For example, PMPA-MD determined a hydrolysis barrier ($E_{\text{a}}^{\text{hydro}}$) of 0.77 eV in the *cis*-2Cl-Hydro. step, agreeing well with the result of 0.70 eV via constrained MD [Fig. S3(b)]. By contrast, MPA-MD exhibits insufficient accuracy in handling H_2PtCl_6 hydrolysis

reactions with noticeable energy deviation of 0.3–0.5 eV (approximately the interaction energy of a hydrogen bond^{42,43}) between $\Delta E_{\text{Ambient}}$ and $\Delta E_{\text{Projective}}$. Intriguingly, we noticed a positive deviation in $\Delta E_{\text{Ambient}}$ for the process involving the transition from separated to merged solvation shells (Fig. 1 and Table I), whereas negative ones for the process with the opposite transition of solvation states (Fig. 2 and Table II). This indicates the underestimation of solvation interactions for merged solvation states by the MPA-MD method, and demonstrates the intrinsic solvation projection nature of hydrogen bond interactions, achieving a unified understanding of the nature of both covalent and hydrogen bonds.

Nevertheless, it should be noted that, for situations where IS and FS remain consistently either merged, or separated solvation shells, MPA-MD can provide comparable results with the benchmark constrained MD. This is particularly suitable for most liquid/solid interface reactions, e.g., the O/O coupling or proton

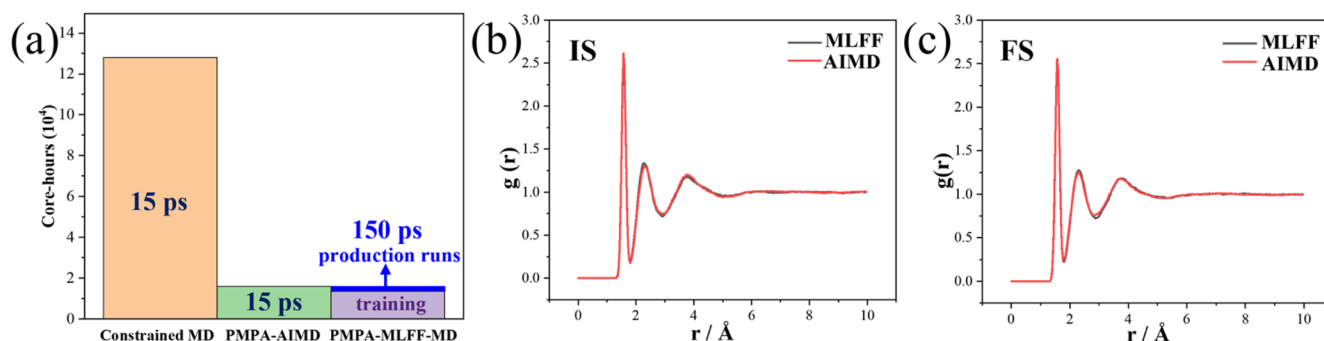


FIG. 3. (a) Statistical computational costs (in core-hours) for calculating one hydrolysis step via the constrained MD and PMPA-MD methods employing AIMD (15 ps for each sample) or MLFF-MD (150 ps for each sample) based simulations. Deviations in H–H pair correlation function (PCF) determined via MLFF-MD and AIMD simulations over a 3 ps trajectory for (b) IS and (c) FS of the *cis*-4Cl-Hydro. process.

transfer at the water/TiO₂ interface [Figs. S4(a) and S4(b)], as demonstrated in our previous works.^{23,24} Even for the ligand exchange reaction in homogeneous catalysis, PMPA-MD also presents reliable accuracy for the formation of a semi-detached ligand (no transition of merged/separated solvation shells). Taking the 1Cl-Hydro. step as an example, the formation of a semi-detached Cl[−] [sharing hydrogen bonds with the Pt complex; IS → FS' in Fig. S4(c)], was calculated to be endothermic 0.17 eV by the PMPA-MD method, showing excellent accuracy relative to the results of constrained MD with slight energy deviations of ~0.03 eV (Table S1).

In addition, Fig. 3(a) shows the computational cost for determining the reaction energy of one elementary step in H₂PtCl₆ hydrolysis (15 ps for each AIMD simulation) using the constrained MD and PMPA-MD (or PMPA-MD) methods. One can see a dramatic decrease in computational cost for PMPA-MD, being ~1/8 of that of the constrained MD method, at comparable calculation accuracy (Tables I and II). Moreover, while the constrained MD requires accurate Helmholtz free-energy gradients $\frac{\partial A}{\partial l}$ and, therefore, explicit AIMD trajectories, PMPA-MD collects structural samples from MD simulations and determines $E_{\text{sol-included}}$ via DFT calculations, thus allowing to get accelerated by performing machine learning force-field^{35,36}-based MD simulations (PMPA-MLFF-MD). Taking the *cis*-4Cl-Hydro. step as an example, although the on-the-fly training took ~13500 core-hours (~90% of total consumption; Fig. 3(a)) to attain reliable MLFFs for IS and FS [as ensured by Figs. 3(b) and 3(c), Table S2 and Fig. S5], subsequent MLFF-MD simulations can be easily extended to a large time scale at very cheap cost (a 150 ps MLFF-MD consumes ~10% resources as those for a 15 ps AIMD simulation; Fig. S6), yet credible accuracy ($\Delta E_{\text{Projective}}$ of −0.83 vs −0.76 eV via the PMPA-MLFF-MD and PMPA-AIMD, respectively). Overall, these results demonstrate PMPA-MD to be a very powerful tool that realizes consistent treatment for diverse

liquid-phase scenarios (e.g., liquid/solid interfaces, homogeneous solution reactions) with good accuracy and low cost, as well as being compatible with the prevalent MLFF technique for conducting large-scale (in time or space) liquid-phase simulations.

C. Mechanistic properties of H₂PtCl₆ hydrolysis

Figure 4 depicts the structural evolution and energy profile of the entire H₂PtCl₆ hydrolysis process via the PMPA-AIMD method (for details, see Figs. S3 and S6). Thermodynamically, it was observed that the first hydrolysis step is slightly endothermic by 0.11 eV, whereas the last step is strongly exothermic by −2.54 eV, signifying the structural resistance in decomposing the fully chlorinated or hydrolyzed Pt complex (i.e., the hydration of [PtCl₆]^{2−} or chlorination of [Pt(OH)_x(H₂O)_{6−x}]^{4−x}). Similar behaviors have also been discovered in the hydrolysis of [AuCl₄][−] species.⁴⁴ All the intermediate hydrolysis steps involving mixed −Cl/−H₂O(OH) coordination Pt complex are found to be moderately exothermic by −0.3 to −1.8 eV. Moreover, we determined the hydrolysis barrier ($E_{\text{a}}^{\text{hydro}}$) of three representative steps (1Cl-Hydro., *cis*-2Cl-Hydro., and *cis*-4Cl-Hydro.) to be around 0.5–0.9 eV (Fig. S3), which correlates linearly ($R^2 = 0.86$) with the reaction energy (Fig. S8), validating the effectiveness of Brønsted–Evans–Polanyi (BEP) relationship^{45–47} in liquid-phase reactions. Obviously, the unfavorable energetics of the first hydrolysis step (the largest barrier of 0.86 eV and endothermic reaction energy) dictate the rate-determining step, potentially retarding the entire process. However, it should be noted that these calculations were performed using a limited model with high solution acidity (containing 1 H₂PtCl₆ and 100 H₂O with estimated pH < 0). Hence, one may expect that H₂PtCl₆ hydrolysis is prone to occur under typical experimental conditions using dilute H₂PtCl₆ solutions (pH = 3–4) or K₂PtCl₆ precursors (pH ~ 7).^{48–50} The calculation results are consistent with the reports of Kramer and Koch⁵¹

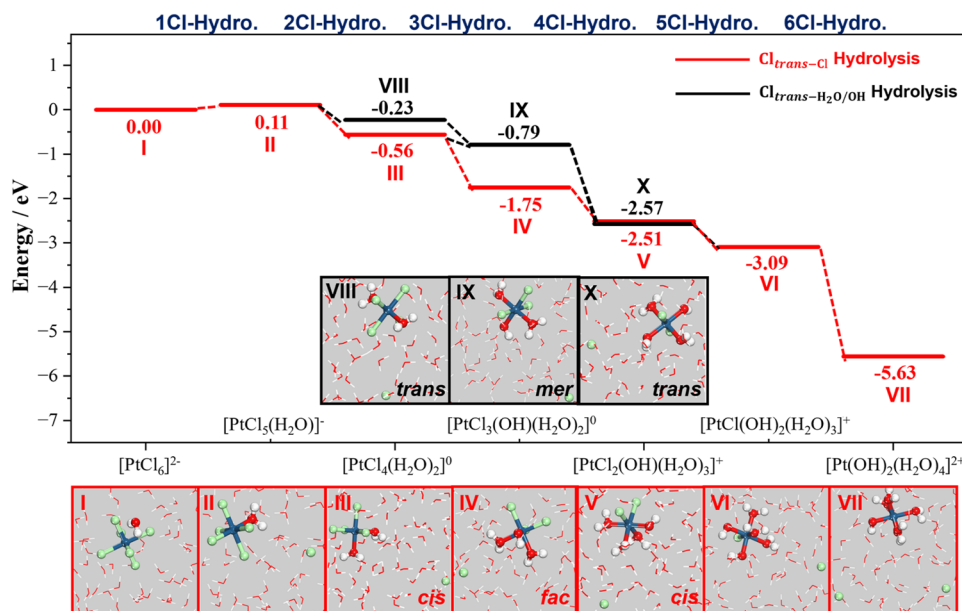


FIG. 4. Calculated energy profile and structural evolution of the H₂PtCl₆ hydrolysis via the PMPA-AIMD method. The stereoselective preference of detaching a Cl[−] in a *trans* configuration to other −Cl ligands (Cl_{trans}−Cl hydrolysis; red curve) over that of −H₂O/OH groups (Cl_{trans}−H₂O/OH hydrolysis; black curve) is observed. The geometric isomers of the intermediates are labeled as *cis*, *trans*, *fac* (facial), and *mer* (meridional).

and Vasilchenko *et al.*,⁵² which demonstrated that H_2PtCl_6 hydrolysis is severely inhibited under strongly acidic conditions ($\text{pH} < 2$) and conversely facilitated as the pH increases.

Geometrically, two major structural issues need to be particularly highlighted. The first one is the stereoselective preference of the detaching Cl^- , taking the 2Cl-Hydro. step as an example, as there are two types of $-\text{Cl}$ ligands holding a *trans* configuration to either an existing $-\text{H}_2\text{O}$ (insert VIII; denoted as $\text{Cl}_{\text{trans}-\text{H}_2\text{O}}$) or other $-\text{Cl}$ (insert III; denoted as $\text{Cl}_{\text{trans}-\text{Cl}}$). We performed the crystal orbital Hamilton population (COHP)⁵³ analysis on the Pt–Cl bond strength of the two types of $-\text{Cl}$ ligands and found that $\text{Cl}_{\text{trans}-\text{Cl}}$ exhibits weaker coordination interaction than the $\text{Cl}_{\text{trans}-\text{H}_2\text{O}/\text{OH}}$ (Fig. S9), signifying the ease of dechlorination of $\text{Cl}_{\text{trans}-\text{Cl}}$ ligands. Consistently, the determined energy profile (Fig. 4) also showed that the $\text{Cl}_{\text{trans}-\text{Cl}}$ pathway (red curve) is more favored in thermodynamics than the $\text{Cl}_{\text{trans}-\text{H}_2\text{O}/\text{OH}}$ type (black curve). Moreover, the stereoselective preference of $\text{Cl}_{\text{trans}-\text{Cl}}$ can be rationally explained by the *trans* effect in metal complex theory,⁵⁴ which states that a ligand with stronger coordination to the metal center (e.g., $-\text{Cl} > -\text{H}_2\text{O}/\text{OH}$)⁵⁵ will weaken the opposite coordination and thereby facilitate the substitution of the ligand *trans* to it. Such *trans* effect has also been observed in the hydrolysis of other halide-coordinated Pt complexes via the ^{195}Pt NMR technique.^{52,56}

The second issue is the existence form of the coordinating water, i.e., molecular $-\text{H}_2\text{O}$ or dissociated $-\text{OH}$. Figure 4 shows the most frequent existence forms of each hydrolyzed intermediate (other candidate structures in Table S3) determined via AIMD simulations, which generally prefer to carry $-\text{OH}$ ligands after detaching $\geq 2\text{Cl}^-$ ions, consistent with experimental ^{195}Pt NMR characterizations.^{6,57} Note that the deprotonation of $-\text{H}_2\text{O}$ ligands depends on the solution acidity. For example, the hydrolysates of

the initial three hydrolysis steps largely preserve molecular $-\text{H}_2\text{O}$ ligands and exist in the form of $[\text{PtCl}_5(\text{H}_2\text{O})]^-$, $[\text{PtCl}_4(\text{H}_2\text{O})_2]^0$, and $[\text{PtCl}_3(\text{OH})(\text{H}_2\text{O})_2]^0$, under strong acidic solution (H_5O_2^+ present in the solution with estimated $\text{pH} \leq 0.27$; inserts II, III, and IV in Fig. 4). However, these complexes are prone to deprotonate into $[\text{PtCl}_5(\text{OH})]^{2-}$, $[\text{PtCl}_4(\text{OH})_2]^{2-}$, and $[\text{PtCl}_3(\text{OH})_2(\text{H}_2\text{O})]^-$ as pH rises to neutral, as demonstrated either by removing HCl out of the model (ISs of 2Cl-Hydro., 3Cl-Hydro., and 4Cl-Hydro. steps in Table S3) or replacing H^+ with Na^+ using the same AIMD approach [Figs. S10(a)–S10(c)]. Intriguingly, the number of $-\text{OH}$ ligands in Pt complexes, including those with primarily aqueous coordination, is limited to a maximum of 2 under acidic or neutral conditions but can exceed 2 in alkaline environments. These are, for instance, the determined structures of $[\text{PtCl}_3(\text{OH})_2(\text{H}_2\text{O})]^-$, $[\text{PtCl}_2(\text{OH})_2(\text{H}_2\text{O})_2]^0$, $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$, and $[\text{Pt}(\text{OH})_2(\text{H}_2\text{O})_4]^{2+}$ in non-alkaline solutions [Fig. S10(c); ISs and FSs of 5Cl-Hydro. and 6Cl-Hydro. in Table S3], whereas $[\text{PtCl}_3(\text{OH})_3]^{2-}$ when introducing OH^- to the model with an estimated pH of 10 via AIMD simulations (Fig. S11). These AIMD results agree with experimental observations that $[\text{Pt}(\text{OH})_6]^{2-}$ occurs exclusively under alkaline conditions.^{6,52}

The determined hydrolysis intermediates and pH-dependent trends enabled us to deduce a broad speciation diagram of H_2PtCl_6 hydrolysis under experimental conditions, as collected in Fig. 5. In general, under acidic conditions, H_2PtCl_6 hydrolysis remains rather suppressed,^{6,51} yielding predominantly anionic species such as $[\text{PtCl}_5(\text{H}_2\text{O})]^-$ and $[\text{PtCl}_4(\text{OH})(\text{H}_2\text{O})]^-$. As solution pH increases to a neutral level, the degree of hydrolysis gradually intensifies and produces mainly cationic Pt complexes such as $[\text{PtCl}_2(\text{OH})(\text{H}_2\text{O})_3]^+$ and $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$. Under alkaline conditions, progressive deprotonation of $-\text{H}_2\text{O}$ ligands turns the hydrolysate

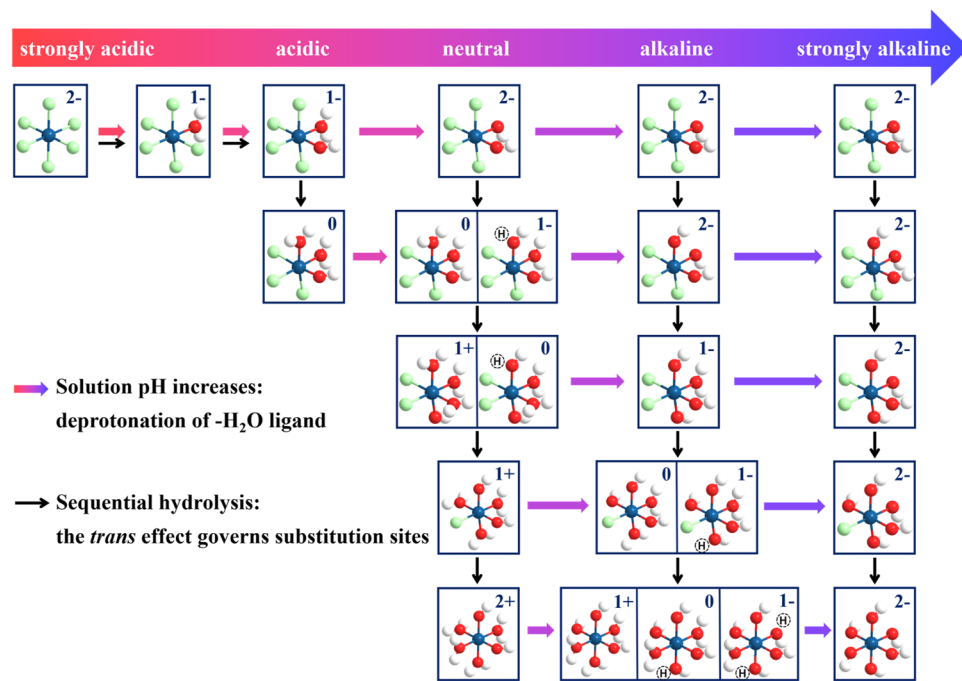


FIG. 5. Deduced speciation diagram of Pt complexes along with the degree of H_2PtCl_6 hydrolysis (indicated by black arrows) and deprotonation events (colored arrows) as induced by varying solution pH (strongly acidic, acidic, neutral, alkaline, and strongly alkaline solutions). The charge states of Pt complexes are labeled in the top-right corner, and the detached proton is indicated by dashed circles.

back into anionic species and ultimately even produces $[\text{Pt}(\text{OH})_6]^{2-}$ under strong alkaline conditions, consistent with experimental observations.⁵²

D. Insights into pH-modulated Pt deposition

Having identified pH-dependent speciation of Pt complexes in solutions, we are stimulated to shed light on a more complicated matter of controlling deposited Pt morphology (particles vs single atoms) in experimental synthesis.^{6,8} First, we investigated the adsorption properties of representative Pt complexes on a model surface of anatase $\text{TiO}_2(101)$, given that the as-formed Pt/ TiO_2 composites are widely used in photocatalysis. The following features were observed: (i) Pt complexes preferentially anchor at surface Ti_{5c} sites through the bidentate over monodentate configuration [Fig. S12(a)]; (ii) the $-\text{OH}$ ligand exhibits stronger adsorption strength than the $-\text{Cl}$ ligand [Figs. S12(b) and S12(c)]; (iii) anionic and cationic complexes, such as $[\text{PtCl}_3(\text{OH})_2(\text{H}_2\text{O})]^-$ and $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$ in Figs. 6(a) and S13(a)–S13(c), are inclined to adsorb on protonated (H^+ -rich and positively charged) and hydroxylated (OH^- -rich and negatively charged) surfaces, respectively, owing to the electrostatic interactions,⁵⁸ whereas charge-neutral species (e.g., $[\text{PtCl}_2(\text{OH})_2(\text{H}_2\text{O})_2]^0$) shows no adsorption preference. Accordingly, one may imagine that the quantity of surface-adsorbed Pt complexes rises as the solution pH gradually increases, because of the greater degree of hydrolysis (thus more $-\text{OH}$ ligands) and/or matched charge states between Pt complexes and TiO_2

surface (Fig. 5). However, it should be noted that, owing to the competitive occupation of surface Ti_{5c} sites against OH^- ions, the adsorption of Pt complexes on TiO_2 surface would be prohibitively suppressed under strongly alkaline conditions with condensed OH^- ions.

Subsequently, since the formation of deposited Pt particles requires the reduction of Pt(IV) complexes into Pt(0), we investigated the primary reduction of Pt(IV) into Pt(II) species under diverse surface conditions [Figs. 6(b) and S11(d)]. Structure optimization and Bader charge analyses confirmed that Pt(IV) complex typically forms an octahedral coordination center, whereas Pt(II) favors a square-planar geometry and, thus, loses two ligands. Such structural characteristics are consistent with the coordination chemistry knowledge and experimental determinations of the coordination structures of M(IV) and M(II) complexes (e.g., $\text{M} = \text{Pt}$ and Pd)^{59,60} Intriguingly, when one of the bidentate anchoring ligands ($-\text{Cl}$ or $-\text{OH}$) is in a *trans* configuration to a $-\text{H}_2\text{O}$ ligand, itself and the opposing $-\text{H}_2\text{O}$ are more likely to break away concurrently (denoted as *trans*- $\text{OH}/\text{H}_2\text{O}$ or *trans*- $\text{Cl}/\text{H}_2\text{O}$; Fig. S14). Moreover, it was found that Pt(IV) reduction exhibits significant sensitivity to surface environments. Taking the $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$ hydrolysate featuring *trans*- $\text{OH}/\text{H}_2\text{O}$ as an example, the reduction energy (E_{red}) was determined to be endothermic by 0.34, 2.19, and 2.24 eV on the H^+ -rich, charge-neutral (PZC), and OH^- -rich TiO_2 surfaces, respectively [Fig. 6(b)]. Even under photocatalytic conditions with excess electrons, Pt(IV) reduction on OH^- -rich TiO_2 surface remains to be the least exothermic process ($E_{\text{red}} = -0.60$ eV; red

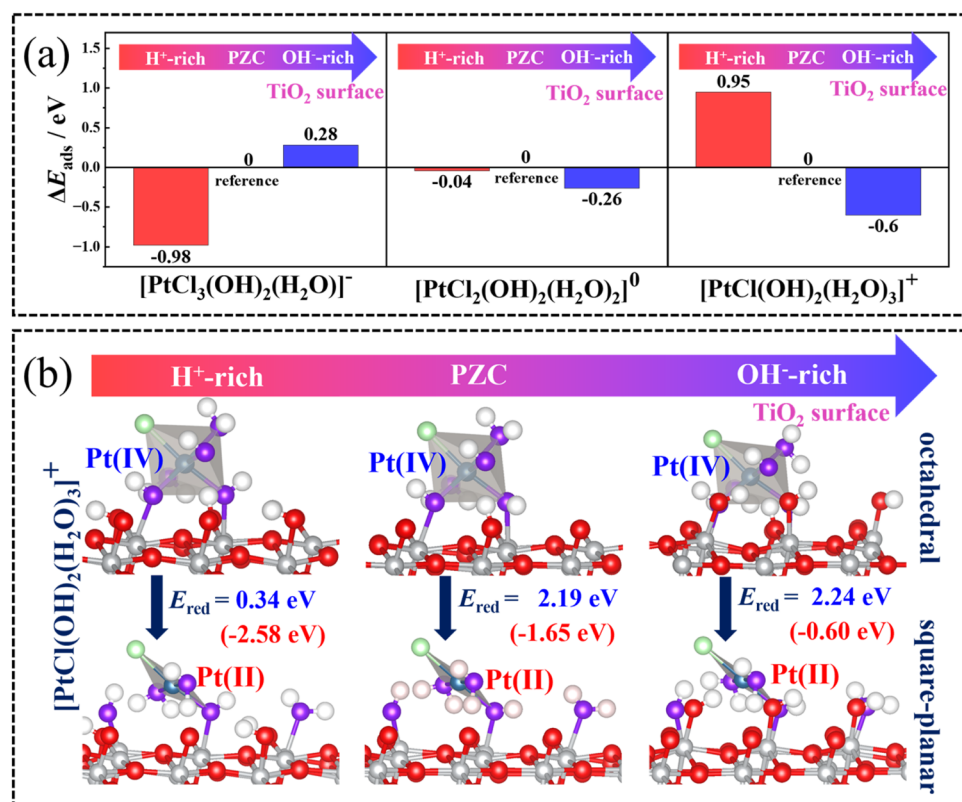


FIG. 6. Calculated (a) relative adsorption energy (ΔE_{ads}) of the anionic $[\text{PtCl}_3(\text{OH})_2(\text{H}_2\text{O})]^-$, charge-neutral $[\text{PtCl}_2(\text{OH})_2(\text{H}_2\text{O})_2]^0$, and cationic $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$, and (b) structures and the Pt(IV) $\rightarrow$ Pt(II) reduction energy (E_{red}) of a representative $[\text{PtCl}(\text{OH})_2(\text{H}_2\text{O})_3]^+$ hydrolysate, on H^+ -rich, charge-neutral (PZC) and OH^- -rich anatase $\text{TiO}_2(101)$ surfaces. E_{red} values in blue and red represent the situations of thermocatalytic and photocatalytic (with excess electrons) reduction, respectively. O atoms originated from Pt complexes are presented in purple.

values in brackets). This is because OH^- -rich surface promotes the dissociation of $-\text{H}_2\text{O}$ ligand (forming $-\text{OH}$ ligand via deprotonation) and stabilizes the octahedral Pt(IV) complexes, whereas H^+ -rich surface hinders the dissociation (preserves *trans*- $\text{OH}/\text{H}_2\text{O}$ or *trans*- $\text{Cl}/\text{H}_2\text{O}$ configuration) and, thus, facilitates the detachment of $-\text{H}_2\text{O}$ ligand and formation of square-planar Pt(II) species.

Finally, as the adsorption and subsequent reduction of Pt(IV) complexes on the TiO_2 surface have been demonstrated to be controllable by altering pH conditions, we are able to clarify some prevalent but puzzling experimental observations on Pt deposition.⁸ Under strongly acidic conditions ($\text{pH} \sim 0$), $[\text{PtCl}_6]^{2-}$ hydrolysis is significantly hampered by endothermic energetics ($\text{I} \rightarrow \text{II}$ in Fig. 4), resulting in unfavorable adsorption and reduction of $[\text{PtCl}_6]^{2-}$ on the TiO_2 surface (Fig. S12), and thus, no obvious deposition of Pt particles/clusters was detected. Under moderate acidic conditions ($\text{pH} \sim 4$), incipient $[\text{PtCl}_6]^{2-}$ hydrolysis produces anionic Pt(IV) hydrolysates with *trans*- $\text{Cl}/\text{H}_2\text{O}$ or *trans*- $\text{OH}/\text{H}_2\text{O}$ ligands (Fig. 5), which, along with their enhanced adsorption [Fig. 6(a)] and reduction [Figs. 6(b) and S14] on the H^+ -rich TiO_2 surface, rationally accounts for the formation of agglomerated Pt particles. As pH approaches the PZC of anatase TiO_2 (6.2–8.4),^{8,61,62} the adsorption promotion via electrostatic interactions [H^+ -rich or OH^- -rich surfaces; Fig. 6(a)] disappears, although an enhanced degree of $[\text{PtCl}_6]^{2-}$ hydrolysis, thus leading to uniformly distributed Pt precursors and eventually Pt single atoms rather than agglomerated particles on the surface. Under strongly alkaline conditions ($\text{pH} \sim 12$), deposition of Pt on TiO_2 is significantly limited, owing to (i) the competition against condensed OH^- ions for adsorption sites and (ii) suppressed Pt(IV) reduction on OH^- -rich surface [Fig. 6(b)]. Overall, by developing the robust PMPA-MD method and clarifying the speciation mechanism of $[\text{PtCl}_6]^{2-}$, this study provides a novel perspective on the projection nature of solvation interactions and a clear rationale for the complicated experimental observation of pH-dependent morphology control of Pt depositions (see Fig. 3 in Ref. 8).

IV. CONCLUSION

This work unveiled the projection nature of solvation interactions and developed a robust and efficient PMPA-MD method for liquid-phase reaction simulations, based on which it clarified the reaction mechanism as well as pH-dependent speciation of H_2PtCl_6 hydrolysis. Major results are summarized as follows:

- I. By projecting intermolecular solvation interactions onto each atom/species, the projective solvation approach presents a consistent and reliable calculation of diverse solution scenarios involving transitions between merged and separated solvation shells.
- II. The PMPA-MD method successfully elucidated the entire H_2PtCl_6 hydrolysis mechanism with high accuracy (energy deviations within 0.1 eV) at a computational cost nearly an order of magnitude lower than that of constrained MD simulations.
- III. H_2PtCl_6 hydrolysis is generally an exothermic process with surmountable barriers obeying the BEP relationship, where the dechlorination stereoselectivity is governed by the *trans* effect and substituent ligand ($-\text{H}_2\text{O}/-\text{OH}$) being pH-dependent.

- IV. The speciation of Pt(IV) hydrolysates determines their bidentate adsorption configuration and subsequent reduction into square-planar Pt(II) complexes on TiO_2 surface, thereby rationalizing, together with the adaptive surface structures (H^+ -rich, charge-neutral, OH^- -rich), the pH-dependent Pt deposition morphology (particles vs single atoms) observed in experiments.

SUPPLEMENTARY MATERIAL

The supplementary material for this paper is available online. AIMD trajectories, free-energy gradient profiles and representative structures of H_2PtCl_6 hydrolysis; linear BEP relationship of H_2PtCl_6 hydrolysis; COHP analyses on the coordination bond strength; adsorption structures of diverse Pt complexes on TiO_2 surfaces.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Fei Li: Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Visualization (equal); Writing – original draft (equal). **Haosheng Niu:** Formal analysis (equal); Software (equal). **Meiyang Wang:** Formal analysis (equal). **Xuezhi Duan:** Formal analysis (equal). **Dong Wang:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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